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CERTIFIED STRONG-COUPLING ENDPOINT AND SHORT-RANGE ENTANGLEMENT FOR THE SU(N) KOGUT–SUSSKIND VACUUM

Main result: the strong-coupling threshold is reduced to a single certified constant κ_Y, $g_0(N) = (6N^2\kappa_Y/(N^2-1))^{1/4} \le (8\kappa_Y)^{1/4}$ for every N, with the link-embedding and boundary estimates proved; vacuum correlations cluster exponentially with an explicit correlation length, uniformly in volume; and an entanglement area law holds up to a polylogarithmic factor under an explicit energy-controlled locality hypothesis. Certifying κ_Y and removing the logarithm remain open.

Roger Cooperman

https://orcid.org/0009-0009-5011-2135

and A.E. Petrunina V.V.

https://orcid.org/0009-0002-4595-736X

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Abstract

This paper is the strong-coupling companion to “Continuum 4D SU(N) Yang–Mills with a Mass Gap” [4]. It takes the four strong-coupling results of Provenance B8 of the weak-coupling programme [2] and the deliverable of workstream 1 of Provenance B2, a numerical threshold $g_0(N)$, and puts them on a single footing. Theorem 1 proves the link-embedding and boundary estimates required by Yarotsky's gap theorem and reduces $g_0(N)$ to one constant κ_Y of that theorem, uniformly in N. Lemma 2 is a Lieb–Robinson bound with explicit constants despite unbounded electric terms. Theorem 3 proves exponential clustering of vacuum correlations, uniformly in volume, by adapting the Nachtergaele–Sims argument to unbounded on-site operators. Lemma 4 bounds the electric energy of every link by $O(g^{-4})$ and gives a volume-law entropy bound. Theorem 5 proves an entanglement area law up to a factor $(1 + \ln|A|)^3$ under an explicit energy-controlled locality hypothesis, and an exact area law under a further entangling-rate hypothesis. As in the companion papers, every definition, statement and proof is written in mathematical symbols only; the discussion is confined to this abstract, the provenance, the status section and the references.

Provenance

Here	Origin	Status before this paper
Theorem 1: certified endpoint	B2, workstream 1; gate $\mathfrak{G}_1$ of [4, Lemma 1]; A1 of [2]	Open
Lemma 2: Lieb–Robinson bound	B8; [2, Lemma 5.1]	Proved, constants implicit
Theorem 3: exponential clustering	B8; [2, Corollary 5.2]	Tentative
Lemma 4: local electric energy	B8; [2, Lemma 5.3]	Proved
Theorem 5: entanglement area law	B8; [2, Proposition 5.4]	Tentative, three gaps

Theorem 1 discharges the embedding part of gate $\mathfrak{G}_1$ in [4], used by Lemma 1(i), by the endpoint hypothesis Y of Lemma 4 and by Lemma 5(i) for $g \geq g_0$. The single remaining input is the numerical value of κ_Y.

0 Definitions

D1 $N \in \mathbb{N}_{\geq 2}$, $G := SU(N)$, $\langle X, Y\rangle := -\mathrm{Tr}(XY)$ on $\mathfrak{su}(N)$, $D := \dim G = N^2 - 1$, $c_N := (N^2-1)/N$, $\rho :=$ fundamental representation.

D2 $\Lambda_L := \{0, \ldots, L\}^3$, $E_L := \{(x, i) : x, x + e_i \in \Lambda_L\}$, $P_L := \{(x, i, j) : i < j, x + e_i + e_j \in \Lambda_L\}$, $\partial(x, i, j) := ((x, i), (x + e_i, j), (x + e_j, i)^{-1}, (x, j)^{-1})$, $U_p := \prod_{e \in \partial p} U_e$.

D3 $\mathscr{H}_L := \otimes_{e \in E_L} L^2(G, dU)$, $h_e := -c_N^{-1} \Delta_e$, $\psi_p := -(2/c_N)\,\mathrm{ReTr}\,\rho(U_p)$, $\lambda := g^{-4}$, $\Omega_0 := \mathbb{1} \in \mathscr{H}_L$.

D4 $\bar{H}_L := \Sigma_{e \in E_L} h_e + \lambda \Sigma_{p \in P_L} \psi_p$, $H_L(g) := (g^2/2)\,\Sigma_e(-\Delta_e) + g^{-2}\,\Sigma_p[N - \mathrm{ReTr}\,\rho(U_p)] = (g^2 c_N/2)\,\bar{H}_L + g^{-2} N |P_L|\,\mathbb{1}$.

D5 $E_0 := \inf \mathrm{spec}\,\bar{H}_L$, $\Omega_L : \bar{H}_L \Omega_L = E_0 \Omega_L$, $\|\Omega_L\| = 1$, $P_{0,L} := |\Omega_L\rangle\langle\Omega_L|$, $\gamma_L := \inf(\mathrm{spec}\,\bar{H}_L \setminus \{E_0\}) - E_0$, $\langle\cdot\rangle_L := \langle\Omega_L, \cdot\,\Omega_L\rangle$.

D6 $\iota : E_L \to \mathbb{Z}^3$, $\iota(x, i) := 2x + e_i$; $\Lambda_0 := \{0, 1, 2\}^3$; $B_L := \{0, \ldots, 2L + 2\}^3$; $\varphi_{2x} := \lambda \Sigma_{(x,i,j) \in P_L} \psi_{(x,i,j)}$, $\varphi_z := 0$ $(z \notin 2\mathbb{Z}^3)$.

D7 $\mathscr{Y}(\beta) := \{ \Sigma_{z \in \Lambda} h_z + \Sigma_{z + \Lambda_0 \subset \Lambda} \varphi_z : h_z = h_z^* \geq 0, \dim\ker h_z = 1, \mathrm{spec}\,h_z \cap (0, 1) = \varnothing, \varphi_z = \varphi_z^* \in \mathscr{A}_{z+\Lambda_0}, \sup_z \|\varphi_z\| \leq \beta \}$ [3, §1].

D8 $(c_1, c_2) \in \mathbb{R}_{>0}^2 : \forall \beta < c_1\ \forall \Lambda \subset \mathbb{Z}^3$ finite $\forall H \in \mathscr{Y}(\beta) : \dim\ker(H - \inf\mathrm{spec}\,H) = 1 \wedge \mathrm{gap}(H) \geq 1 - c_2\beta$ [3, Thm 1]; $\kappa_Y := \max\{2/c_1, 2c_2\}$, $\beta^* := \kappa_Y^{-1} = \min\{c_1/2, (2c_2)^{-1}\}$.

D9 $g_0(N) := (6N\,\kappa_Y / c_N)^{1/4} = (6N^2\,\kappa_Y / (N^2 - 1))^{1/4}$.

D10 $d(z, z') := \|z - z'\|_\infty$; $\mathscr{A}_X := \mathscr{B}(\otimes_{z \in X} L^2(G)) \otimes \mathbb{1}$; $\|\Phi\|_\mu := \sup_x \Sigma_{z\,:\,x \in z + \Lambda_0} \|\varphi_z\|\, e^{\mu\,\mathrm{diam}(z+\Lambda_0)}$; $\tau_t(A) := e^{it\bar{H}_L} A\, e^{-it\bar{H}_L}$, $\tau^0{}_t(A) := e^{itH_0} A\, e^{-itH_0}$, $H_0 := \Sigma_e h_e$.

D11 $e_4(N, \lambda) := 16N\lambda/c_N + 256\,N(N^2 - 1)\,\lambda^2 / c_N^3$; $Z(\beta) := \mathrm{Tr}_{L^2(G)}\, e^{-\beta h} = \Sigma_{\pi \in \hat{G}}\, d_\pi^2\, e^{-\beta C(\pi)/c_N}$; $s(E) := \inf_{\beta > 0}[\beta E + \ln Z(\beta)]$; $c_Z : \forall \beta \in (0, 1] : \ln Z(\beta) \leq (D/2) \ln \beta^{-1} + c_Z$ [10].

D12 $A \subset E_L$, $A^c := E_L \setminus A$, $\rho_A := \mathrm{Tr}_{A^c} |\Omega_L\rangle\langle\Omega_L|$, $S(\rho) := -\mathrm{Tr}\,\rho \ln \rho$, $h(\varepsilon) := -\varepsilon \ln \varepsilon - (1 - \varepsilon)\ln(1 - \varepsilon)$; $\partial A := \{e \in A : \exists e' \in A^c, d(\iota(e), \iota(e')) \leq 2\}$, $S_w(A) := \{e \in E_L : d(\iota(e), \iota(\partial A)) \leq w\}$.

D13 $\mathfrak{RI}_E(\ell, C_{\mathfrak{RI}}, E_{\mathfrak{RI}}) :\Longleftrightarrow \forall L\ \forall A\ \forall w \in \mathbb{N}\ \exists V_A \in \mathscr{U}(\mathscr{A}_{\iota(A)}), V_B \in \mathscr{U}(\mathscr{A}_{\iota(A^c)}), W \in \mathscr{U}(\mathscr{A}_{\iota(S_w(A))}), \sigma := (V_A \otimes V_B) W \Omega_0 : \|\Omega_L - \sigma\| \leq C_{\mathfrak{RI}} |\partial A|\, e^{-w/\ell} \wedge \sup_e \max\{\langle\sigma, h_e\sigma\rangle, \langle W\Omega_0, h_e W\Omega_0\rangle\} \leq E_{\mathfrak{RI}}$.

D14 $\bar{H}(s) := H_0 + s\lambda \Sigma_p \psi_p$, $s \in [0, 1]$; $\Omega(s) :=$ ground state of $\bar{H}(s)$; $\mathfrak{S}(c_{\mathfrak{S}}) :\Longleftrightarrow \forall L\ \forall A : s \mapsto S(\rho_A(s)) \in \mathrm{AC}[0, 1] \wedge |\partial_s S(\rho_A(s))| \leq c_{\mathfrak{S}} |\partial A|$ a.e.

1 Certified strong-coupling endpoint

Theorem 1.

(i) $\forall g > 0\ \forall L \in \mathbb{N}:\ \iota$ injective $\wedge\ \forall p = (x, i, j) : \iota(\partial p) \subset 2x + \Lambda_0 \wedge \sup_z \|\varphi_z\| \le 6N/(g^4 c_N)$

(ii) $g \ge g_0(N) \Rightarrow \forall L:\ \dim\ker(\bar{H}_L - E_0) = 1 \wedge \gamma_L \ge \tfrac{1}{2} \wedge H_L(g) - E_{0,L} \ge (g^2 c_N/4)(\mathbb{1} - P_{0,L}) \wedge \Omega_L \in \mathscr{H}_L^{\mathrm{phys}}$

(iii) $g_0(2) = (8\kappa_Y)^{1/4} \wedge g_0(3) = (27\kappa_Y/4)^{1/4} \wedge N \mapsto g_0(N) \downarrow \wedge \lim_{N\to\infty} g_0(N) = (6\kappa_Y)^{1/4} \wedge \sup_{N\ge 2} g_0(N) = (8\kappa_Y)^{1/4}$

Proof.

(i.1) $\iota(x, i)_k \equiv \delta_{ik} \pmod 2 \Rightarrow (\iota(x, i) = \iota(x', i') \Rightarrow i = i' \wedge x = x')$.

(i.2) $\iota(\partial(x, i, j)) = \{2x + e_i, 2x + 2e_i + e_j, 2x + e_i + 2e_j, 2x + e_j\} \subset 2x + \Lambda_0$.

(i.3) $\sup_{U\in G}|\mathrm{ReTr}\,\rho(U)| = N \Rightarrow \|\psi_p\| = 2N/c_N \Rightarrow \|\lambda\psi_p\| = 2N/(g^4 c_N)$.

(i.4) $\#\{(i, j) : 1 \le i < j \le 3\} = 3 \Rightarrow \forall x : \#\{p \in P_L : p = (x, \cdot, \cdot)\} \le 3 \Rightarrow \|\varphi_{2x}\| \le 6N/(g^4 c_N)$ ($\partial\Lambda_L$ included). ∎$_{(i)}$

(ii.1) $g \ge g_0(N) \Leftrightarrow g^4 \ge 6N\kappa_Y/c_N \Leftrightarrow 6N/(g^4 c_N) \le \beta^*$.

(ii.2) $\beta^* \le c_1/2 < c_1 \wedge 1 - c_2\beta^* \ge 1 - \tfrac{1}{2} = \tfrac{1}{2}$.

(ii.3) $\iota(E_L) \subset \{0, \ldots, 2L\}^3 \Rightarrow \forall z : \varphi_z \ne 0 \Rightarrow z + \Lambda_0 \subset B_L$; $H^{\mathrm{enl}} := \Sigma_{z\in B_L} h_z + \Sigma_z \varphi_z \in \mathscr{Y}(\beta^*)$ (i, ii.1).

(ii.4) [3, Thm 1] ∘ (ii.2–3) $\Rightarrow \dim\ker(H^{\mathrm{enl}} - \inf\mathrm{spec}\, H^{\mathrm{enl}}) = 1 \wedge \mathrm{gap}(H^{\mathrm{enl}}) \ge \tfrac{1}{2}$.

(ii.5) $R := B_L \setminus \iota(E_L) \Rightarrow H^{\mathrm{enl}} = \bar{H}_L \otimes \mathbb{1} + \mathbb{1} \otimes \Sigma_{z\in R} h_z \Rightarrow \mathrm{gap}(H^{\mathrm{enl}}) = \min\{\gamma_L, 1\} \wedge \dim\ker(\bar{H}_L - E_0) = 1 \Rightarrow \gamma_L \ge \tfrac{1}{2}$.

(ii.6) $H_L(g) = (g^2 c_N/2)\bar{H}_L + \mathrm{const}\cdot\mathbb{1} \Rightarrow H_L(g) - E_{0,L} \ge (g^2 c_N/2)\cdot\tfrac{1}{2}\cdot(\mathbb{1} - P_{0,L})$.

(ii.7) $e^{-t\bar{H}_L}$ positivity improving $\Rightarrow \Omega_L > 0$; $\forall h \in G^{\Lambda_L} : [\mathscr{U}(h), \bar{H}_L] = 0 \wedge \mathscr{U}(h)\Omega_L > 0 \Rightarrow \mathscr{U}(h)\Omega_L = \Omega_L \Rightarrow \Omega_L \in \mathscr{H}_L^{\mathrm{phys}}$. ∎$_{(ii)}$

(iii.1) $6N/c_N = 6N^2/(N^2 - 1) \Rightarrow N = 2 : 8,\quad N = 3 : 27/4$.

(iii.2) $\partial_N[N^2/(N^2 - 1)] = -2N/(N^2 - 1)^2 < 0 \wedge \lim_{N\to\infty} N^2/(N^2 - 1) = 1 \Rightarrow \downarrow \wedge \lim = (6\kappa_Y)^{1/4} \wedge \sup = g_0(2)$. ∎

2 Lieb–Robinson bound

Lemma 2.

$\forall \mu > 0\ \forall X, Y \subset \mathbb{Z}^3$ finite $\forall A \in \mathcal{A}_X\ \forall B \in \mathcal{A}_Y\ \forall t \in \mathbb{R}$:

$\|[\tau_t(A), B]\| \le 2\|A\|\|B\| \min\{ \Sigma_{x\in X} e^{-\mu\, d(x,Y)}, \Sigma_{y\in Y} e^{-\mu\, d(X,y)} \} (e^{2\|\Phi\|_\mu |t|} - 1)$,

$\|\Phi\|_\mu \le 8\, e^{2\mu} \cdot 6N/(g^4 c_N)$

Proof.

(1) $\|\lambda \Sigma \psi_p\| < \infty \wedge H_0 = H_0^* \Rightarrow \bar{H}_L = \bar{H}_L^*$ on $\mathcal{D}(H_0)$ [Kato–Rellich].

(2) $W(t) := e^{-itH_0} e^{it\bar{H}_L}$, $\alpha_t(A) := W(t)AW(t)^* \Rightarrow \tau_t = \tau^0{}_t \circ \alpha_t \wedge \partial_t W(t) = i\, \Sigma_z\, \tau^0{}_{-t}(\varphi_z)\, W(t)$.

(3) $\tau^0{}_t = \otimes_z \mathrm{Ad}\, e^{ith_z} \Rightarrow \forall X : \tau^0{}_t(\mathcal{A}_X) = \mathcal{A}_X \wedge \|\tau^0{}_t(A)\| = \|A\| \Rightarrow \Phi(t) := \{\tau^0{}_{-t}(\varphi_z)\} : \mathrm{supp} = \mathrm{supp}\, \varphi_z \wedge \|\Phi(t)\|_\mu = \|\Phi\|_\mu$.

(4) [6, time-dependent Lieb–Robinson bound] $\circ$ (3) $\Rightarrow \|[\alpha_t(A), B']\| \le 2\|A\|\|B'\|\, \Sigma_{x\in X} e^{-\mu\, d(x,Y)} (e^{2\|\Phi\|_\mu |t|} - 1)$, $B' \in \mathcal{A}_Y$.

(5) $[\tau_t(A), B] = \tau^0{}_t([\alpha_t(A), \tau^0{}_{-t}(B)]) \wedge \tau^0{}_{-t}(B) \in \mathcal{A}_Y \wedge \|\tau^0{}_{-t}(B)\| = \|B\| \Rightarrow$ bound with $\Sigma_{x\in X}$; $[\tau_t(A), B] = -\tau_t([\tau_{-t}(B), A]) \Rightarrow$ bound with $\Sigma_{y\in Y}$.

(6) $\forall x : \#\{z \in 2\mathbb{Z}^3 : x \in z + \Lambda_0\} \le \prod_{k=1}^{3} \#(\{x_k - 2, x_k - 1, x_k\} \cap 2\mathbb{Z}) \le 2^3 \wedge \mathrm{diam}_\infty \Lambda_0 = 2 \wedge$ Thm 1(i) $\Rightarrow \|\Phi\|_\mu \le 8e^{2\mu} \cdot 6N/(g^4 c_N)$. ∎

3 Exponential clustering

Theorem 3.

$g \geq g_0(N) \wedge \mu > 0 \wedge \mu_3 := (\tfrac{1}{2}\,\mu) / (4\|\Phi\|_\mu + \tfrac{1}{2}) \Rightarrow \forall L\ \forall X \cap Y = \varnothing\ \forall A \in \mathscr{A}_X\ \forall B \in \mathscr{A}_Y :$

$|\langle AB\rangle_L - \langle A\rangle_L \langle B\rangle_L| \leq \|A\|\|B\|\,(1 + (2/\pi)\min\{|X|, |Y|\} + (\pi\mu_3\, d(X, Y))^{-1/2})\, e^{-\mu_3\, d(X,Y)}$

$\mu = \tfrac{1}{2} \Rightarrow \mu_3 \geq (\tfrac{1}{4}) / (192e\, N/(g^4 c_N) + \tfrac{1}{2})$

Proof.

(1) Lemma 2 $\Rightarrow \|[\tau_t(A), B]\| \leq 2\|A\|\|B\|\, \min\{|X|, |Y|\}\, e^{-\mu\, d(X,Y)}\, e^{2\|\Phi\|_\mu |t|}$.

(2) Thm 1(ii) $\Rightarrow H := \bar{H}_L - E_0 \geq 0 \wedge \ker H = \mathbb{C}\Omega_L \wedge \operatorname{spec} H \subset \{0\} \cup [\gamma, \infty)$, $\gamma := \tfrac{1}{2}$.

(3) $\tilde{B} := B - \langle B\rangle_L \mathbb{1} \Rightarrow P_{0,L}\tilde{B}\Omega_L = P_{0,L}\tilde{B}^*\Omega_L = 0 \wedge \langle AB\rangle_L - \langle A\rangle_L\langle B\rangle_L = \langle \Omega_L, A\, \tilde{B}\, \Omega_L\rangle \wedge [\tau_t(A), \tilde{B}] = [\tau_t(A), B]$.

(4) $a > 0$, $\tilde{B}(a) := (a/\pi)^{1/2} \int_{\mathbb{R}} \tau_s(\tilde{B})\, e^{-as^2}\, ds \Rightarrow z \mapsto \tau_z(\tilde{B}(a)) := (a/\pi)^{1/2} \int_{\mathbb{R}} \tau_s(\tilde{B})\, e^{-a(s-z)^2}\, ds$ entire $\wedge$ $\|\tau_z(\tilde{B}(a))\| \leq \|\tilde{B}\|\, e^{a(\operatorname{Im} z)^2}$.

(5) (4) $\in \operatorname{span}\{\tau_s : s \in \mathbb{R}\} \Rightarrow$ (4) $\forall H = H^*$ (no $\|H\| < \infty$, no $\dim < \infty$).

(6) [5, Lemma 1 $\wedge$ proof of Thm 2] $\vdash$ ingredients $\subset$ {(1), (2), (4), spectral theorem for $H = H^* \geq 0$, contour integration, dominated convergence}.

(7) (1) $\wedge$ (2) $\wedge$ (4) $\wedge$ (5) $\Rightarrow$ ingredients of (6) ✓ for $\bar{H}_L \Rightarrow$ [5, Thm 2]$|_{b=0}$ with $\lambda \mapsto \mu$, $\|\Phi\|_\lambda \mapsto \|\Phi\|_\mu$, $\gamma \mapsto \tfrac{1}{2}$, $|Y| \mapsto \min\{|X|, |Y|\} \Rightarrow$ claim, $\mu_3 = (\tfrac{1}{2}\mu)/(4\|\Phi\|_\mu + \tfrac{1}{2})$.

(8) $\mu = \tfrac{1}{2} \wedge$ Lemma 2 $\Rightarrow 4\|\Phi\|_{1/2} \leq 4 \cdot 8e \cdot 6N/(g^4 c_N) = 192eN/(g^4 c_N) \Rightarrow \mu_3 \geq \tfrac{1}{4} / (192eN/(g^4 c_N) + \tfrac{1}{2})$. ∎

4 Local electric energy

Lemma 4.

$g \geq g_0(N) \Rightarrow \forall L\ \forall e \in E_L :\ \langle h_e \rangle_L \leq e_4(N, \lambda)$

$\forall \rho : \mathrm{Tr}\, \rho(\Sigma_{e \in A}\, h_e) \leq E \Rightarrow S(\mathrm{Tr}_{A^c} \rho) \leq \inf_{\beta > 0}[\beta E + |A| \ln Z(\beta)];\ \ \therefore S(\rho_A) \leq |A|\, s(e_4)$

Proof.

(1) $\bar{H}_{\setminus e} := \bar{H}_L - h_e \Rightarrow \langle h_e \rangle_L = E_0 - \langle \Omega_L, \bar{H}_{\setminus e} \Omega_L \rangle \leq E_0(\bar{H}_L) - E_0(\bar{H}_{\setminus e})$.

(2) $\forall f \in C(G) : [\bar{H}_{\setminus e}, f(U_e)] = 0 \Rightarrow \bar{H}_{\setminus e} = \int_G^{\oplus} K(u)\, du \Rightarrow E_0(\bar{H}_{\setminus e}) = \min_{u \in G} E_0(K(u))$.

(3) Thm 1(i–ii) on $E_L \setminus \{e\} \Rightarrow K(u) \in \mathscr{Y}(\beta^*) \Rightarrow \exists!\ \varphi(u) > 0 \wedge \mathrm{gap}\, K(u) \geq \tfrac{1}{2} \wedge u \mapsto \varphi(u) \in C^{\infty}$ [11].

(4) $\psi(U_e, U') := \varphi(U_e)(U')$, $\|\psi\| = 1 \Rightarrow E_0(\bar{H}_L) \leq \langle \psi, \bar{H}_L \psi \rangle = \int_G E_0(K(u))\, du + c_N^{-1} \int_G \|\nabla_u \varphi(u)\|^2\, du \leq \max_u E_0(K(u)) + c_N^{-1} \sup_u \|\nabla_u \varphi\|^2$.

(5) $\|K(u) - K(u')\| \leq \Sigma_{p \ni e} \lambda \|\psi_p(u) - \psi_p(u')\| \leq 4 \cdot \lambda \cdot 4N/c_N \Rightarrow \max_u E_0(K(u)) - \min_u E_0(K(u)) \leq 16N\lambda / c_N$.

(6) $\partial_X \varphi = -R(u)(\partial_X K)\varphi \wedge \|R(u)\| \leq 2 \wedge |\mathrm{ReTr}(XW)| \leq \|X\|_1 \leq \sqrt{N|X|} \Rightarrow \|\partial_X \varphi\| \leq 2 \cdot 4 \cdot \lambda(2/c_N)\sqrt{N|X|} \Rightarrow \|\nabla_u \varphi\|^2 \leq (N^2 - 1)(16\sqrt{N}\lambda/c_N)^2$.

(7) $(1, 2, 4, 5, 6) \Rightarrow \langle h_e \rangle_L \leq 16N\lambda/c_N + 256N(N^2 - 1)\lambda^2/c_N^3 = e_4$.

(8) $\sigma_\beta := e^{-\beta \Sigma_{e \in A} h_e}/Z(\beta)^{|A|} \wedge S(\rho_A \| \sigma_\beta) \geq 0$ [9] $\Rightarrow S(\rho_A) \leq \beta\, \mathrm{Tr}(\rho_A \Sigma h_e) + |A| \ln Z(\beta)$; (7) $\Rightarrow \mathrm{Tr}(\rho_A \Sigma_{e \in A} h_e) \leq |A| e_4$. ∎

5 Entanglement area law

Theorem 5.

(i) $g \geq g_0(N) \wedge \mathfrak{I}_E(\ell, C_{\mathfrak{I}}, E_{\mathfrak{I}}) \Rightarrow \forall L\ \forall A \subset E_L,\ |A| \geq 2,\ |\partial A| \geq 1$:

$S(\rho_A) \leq (2\lceil \ell \ln|A| \rceil + 1)^3\ |\partial A|\ s(E_{\mathfrak{I}}) + C_{\mathfrak{I}}|\partial A|(D \ln|A| + 2c_Z) + 2\max\{e_4, E_{\mathfrak{I}}\} + \ln 2 + C_{\mathfrak{I}}|\partial A|\ s(e_4)$

$\therefore \exists c_5 = c_5(N, \ell, C_{\mathfrak{I}}, E_{\mathfrak{I}}) : S(\rho_A) \leq c_5\ |\partial A|\ (1 + \ln|A|)^3$

(ii) $\Xi(c_\Xi) \Rightarrow \forall L\ \forall A : S(\rho_A) \leq c_\Xi |\partial A|$

Proof.

(i.0) $w := \lceil \ell \ln|A| \rceil$, $\varepsilon := C_{\mathfrak{I}}|\partial A| e^{-w/\ell} \leq C_{\mathfrak{I}}|\partial A|/|A|$; $\varepsilon > 1 \Rightarrow |A| < C_{\mathfrak{I}}|\partial A| \Rightarrow$ (Lemma 4) $S(\rho_A) \leq |A| s(e_4) < C_{\mathfrak{I}}|\partial A| s(e_4) \Rightarrow$ claim. $\therefore \varepsilon \leq 1$.

(i.1) $\sigma := (V_A \otimes V_B) W\Omega_0$ (D13) $\Rightarrow \sigma_A := \mathrm{Tr}_{A^c}|\sigma\rangle\langle\sigma| = V_A\ \mathrm{Tr}_{A^c}(|W\Omega_0\rangle\langle W\Omega_0|)\ V_A^* \Rightarrow S(\sigma_A) = S(\mathrm{Tr}_{A^c}|W\Omega_0\rangle\langle W\Omega_0|)$.

(i.2) $\Omega_0 = \otimes_e \mathbb{1}_e \wedge \mathrm{supp}\ W \subset \iota(S_w(A)) \Rightarrow W\Omega_0 = (W\Omega_0^{S_w}) \otimes \Omega_0^{E_L \setminus S_w} \Rightarrow S(\mathrm{Tr}_{A^c}|W\Omega_0\rangle\langle W\Omega_0|) = S(\mathrm{Tr}_{S_w \setminus A}|W\Omega_0^{S_w}\rangle\langle W\Omega_0^{S_w}|)$.

(i.3) Lemma 4 (Gibbs bound) on $S_w \cap A \wedge$ D13 $\Rightarrow S(\sigma_A) \leq |S_w(A) \cap A|\ s(E_{\mathfrak{I}})$.

(i.4) $\#\{z \in \mathbb{Z}^3 : d(z, z_0) \leq w\} = (2w + 1)^3 \Rightarrow |S_w(A)| \leq (2w + 1)^3 |\partial A|$.

(i.5) $\frac{1}{2}\|\rho_A - \sigma_A\|_1 \leq \frac{1}{2}\|\ |\Omega_L\rangle\langle\Omega_L| - |\sigma\rangle\langle\sigma|\ \|_1 \leq \|\Omega_L - \sigma\| \leq \varepsilon$.

(i.6) Lemma 4 $\wedge$ D13 $\Rightarrow \mathrm{Tr}(\rho_A \Sigma_{e \in A} h_e), \mathrm{Tr}(\sigma_A \Sigma_{e \in A} h_e) \leq \mathscr{E} := |A| \max\{e_4, E_{\mathfrak{I}}\}$.

(i.7) [8, Lemma 15] $\Rightarrow |S(\rho_A) - S(\sigma_A)| \leq 2\varepsilon\ S(\gamma(\mathscr{E}/\varepsilon)) + h(\varepsilon)$, $S(\gamma(E')) \leq \beta E' + |A| \ln Z(\beta)\ \forall \beta > 0$.

(i.8) $\beta := |A|^{-1} \leq 1 \wedge$ D11 $\Rightarrow 2\varepsilon S(\gamma(\mathscr{E}/\varepsilon)) \leq 2\mathscr{E}/|A| + 2\varepsilon|A|((D/2) \ln|A| + c_Z) \leq 2\max\{e_4, E_{\mathfrak{I}}\} + C_{\mathfrak{I}}|\partial A|(D \ln|A| + 2c_Z)$; $h(\varepsilon) \leq \ln 2$.

(i.9) $S(\rho_A) \leq S(\sigma_A) + |S(\rho_A) - S(\sigma_A)| \wedge$ (i.3–i.8) $\Rightarrow$ claim; $w \leq \ell \ln|A| + 1 \Rightarrow (2w + 1)^3 \leq (2\ell + 3)^3(1 + \ln|A|)^3 \Rightarrow c_5$. ∎$_{(i)}$

(ii) $\bar{H}(0) = H_0 \Rightarrow \Omega(0) = \Omega_0 = \otimes \mathbb{1}_e \Rightarrow S(\rho_A(0)) = 0$; $\Omega(1) = \Omega_L \wedge \Xi \Rightarrow S(\rho_A) \leq \int_0^1 |\partial_s S(\rho_A(s))|\ ds \leq c_\Xi |\partial A|$. ∎

6 Status

Result	Status	Remaining input
Theorem 1: certified endpoint	Proved	Numerical value of the single constant $\kappa_Y = \max\{2/c_1, 2c_2\}$ from the proof of [3, Thm 1]. The embedding, boundary and N-uniformity estimates are complete.
Lemma 2: Lieb–Robinson bound	Proved	Uses the time-dependent Lieb–Robinson theorem of [6] for bounded interactions.
Theorem 3: exponential clustering	Proved by adaptation	The argument of [5, Thm 2] is used with Lemma 2 in place of its Lieb–Robinson bound and with Gaussian time-smoothing in place of analyticity of bounded dynamics. [5] is stated for finite-dimensional sites; a line-by-line check of the adapted steps (6)–(7) against [5] is recommended before publication.
Lemma 4: local electric energy	Proved	None beyond Theorem 1.
Theorem 5(i): area law up to $(1 + \ln\|A\|)^3$	Conditional	Hypothesis $\mathfrak{A}_E$: a quasi-local decomposition of the vacuum with energy control. Automorphic equivalence [7] supplies the decomposition; the energy bounds on σ and $W\Omega_0$ are open.
Theorem 5(ii): exact area law	Conditional	Hypothesis $\mathfrak{S}$: an energy-constrained small-incremental-entangling bound. Known proofs use finite local dimension.

Compared with Proposition 5.4 of [2], which claimed $S(\rho_A) \le C|\partial A|$ with three open gaps, Theorem 5 makes the cost of infinite-dimensional links explicit: entropy continuity in infinite dimensions requires energy control, and the continuity error, of order $\varepsilon|A| \ln|A|$, forces a strip of width $\ln|A|$. Removing the polylogarithm is equivalent in difficulty to $\mathfrak{S}$.

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GitHub repository: https://github.com/teknoquants/CERTIFIED-STRONG-COUPLING-ENDPOINT-AND-SHORT-RANGE-ENTANGLEMENT-FOR-THE-SU-N-KOGUT-SUSSKIND-VACUUM
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